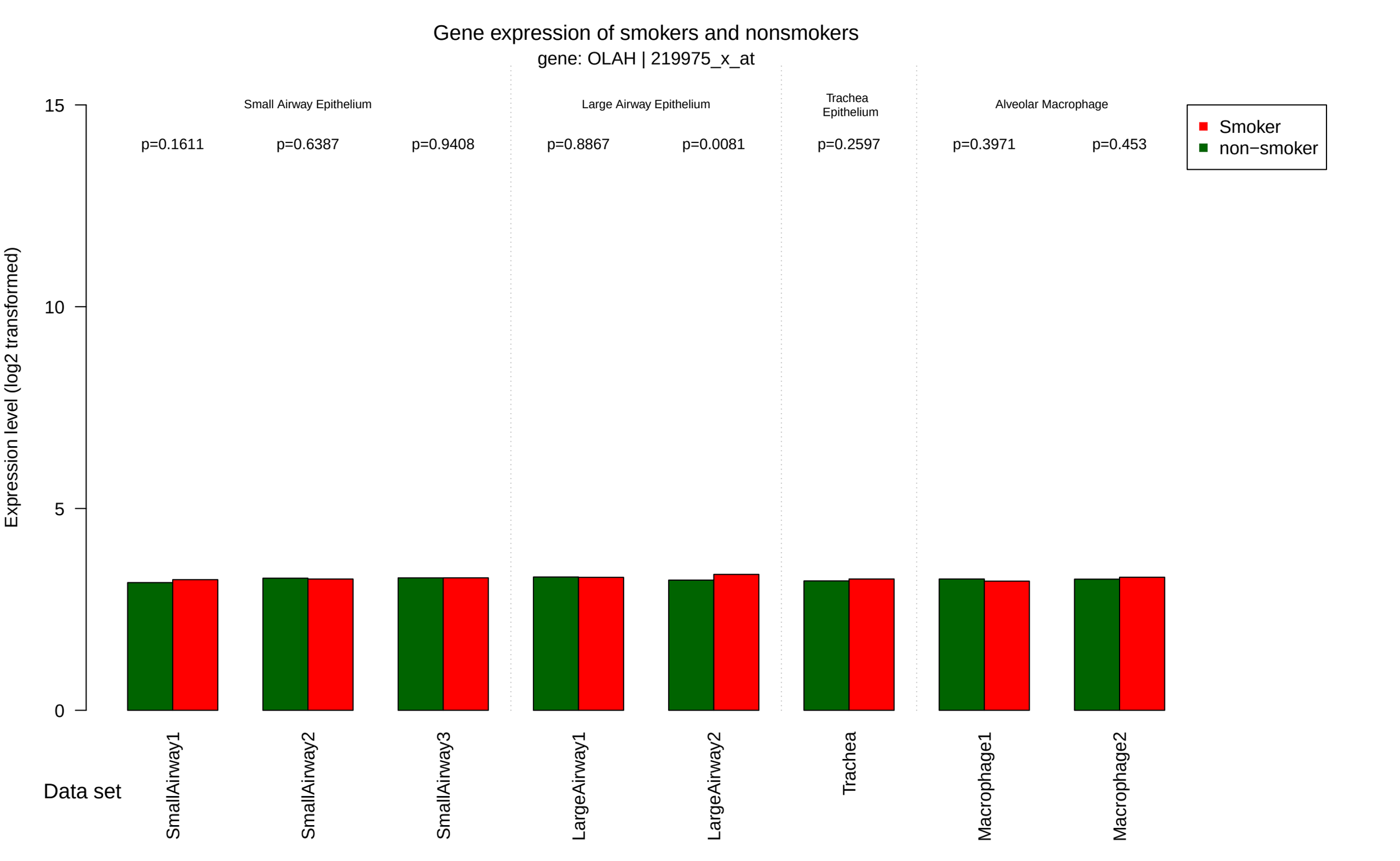
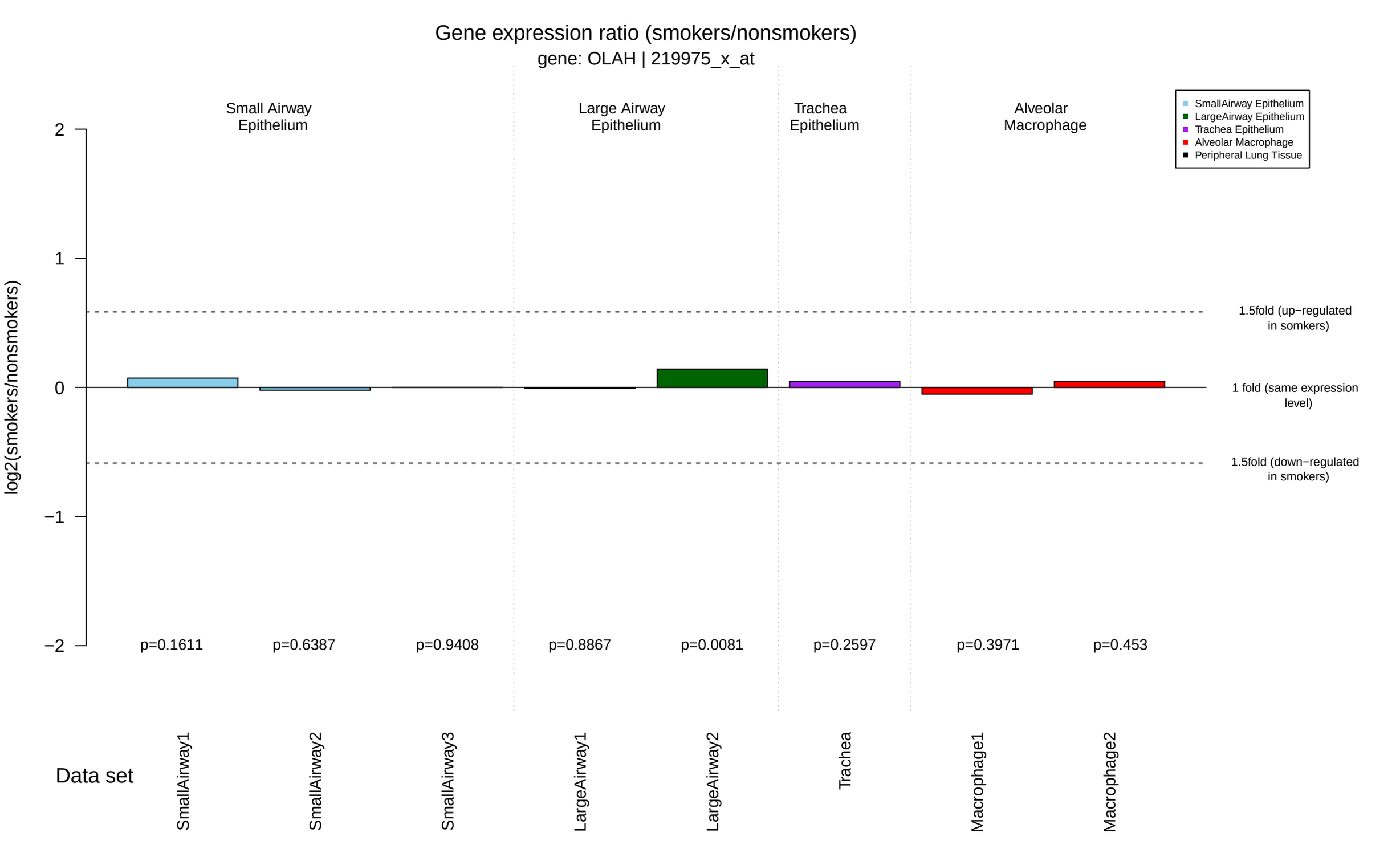
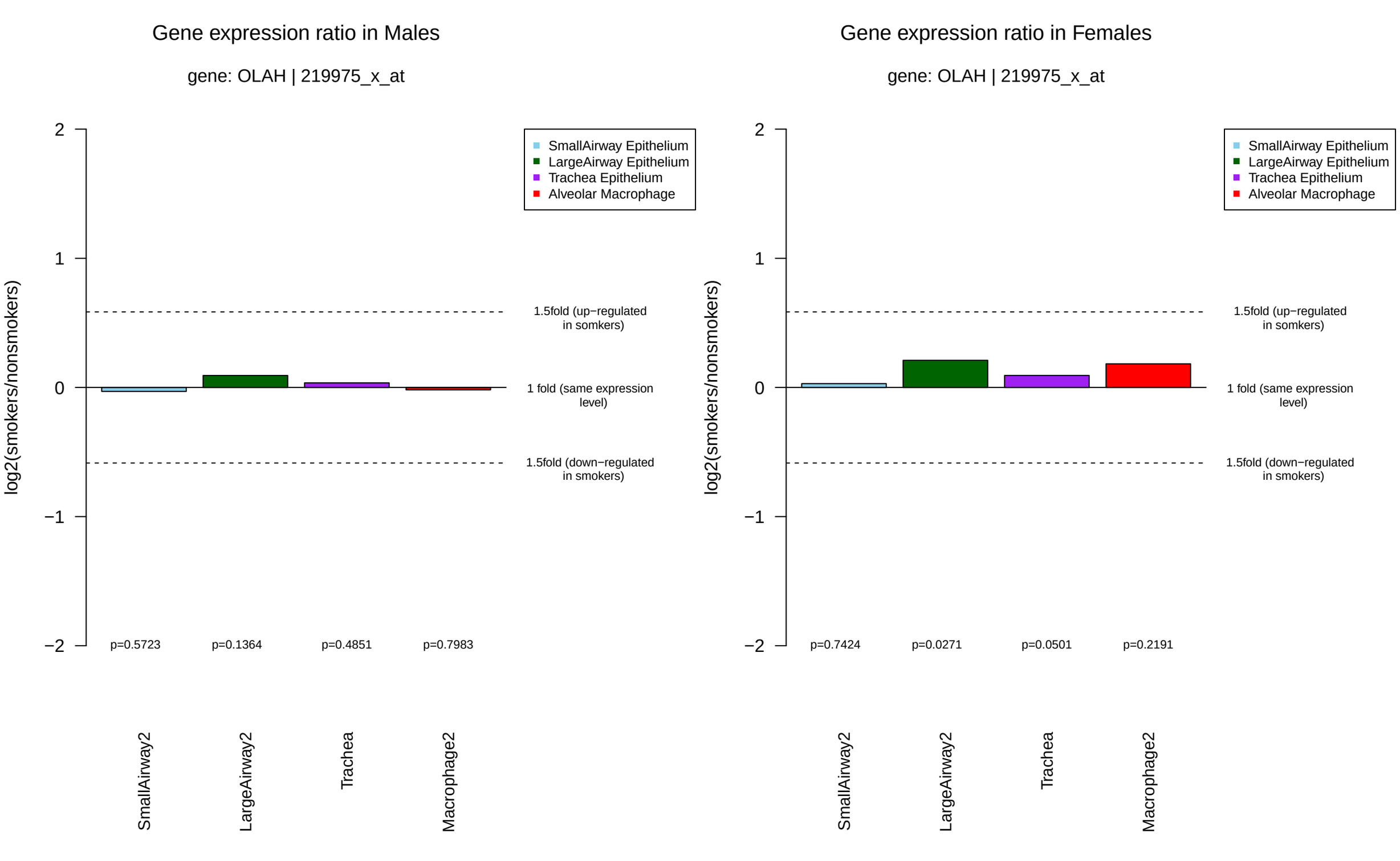
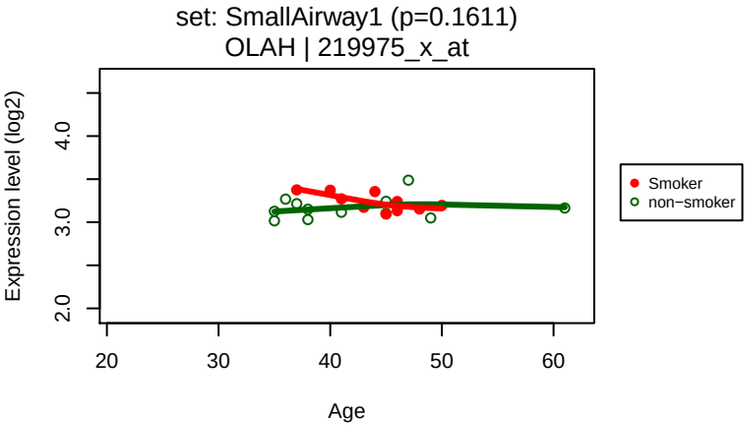
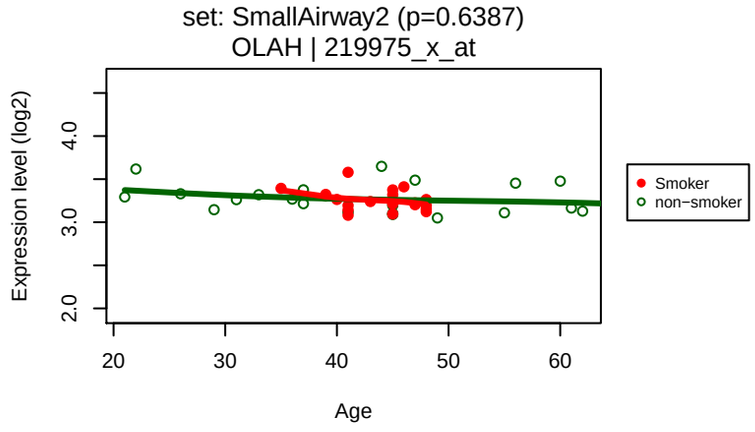
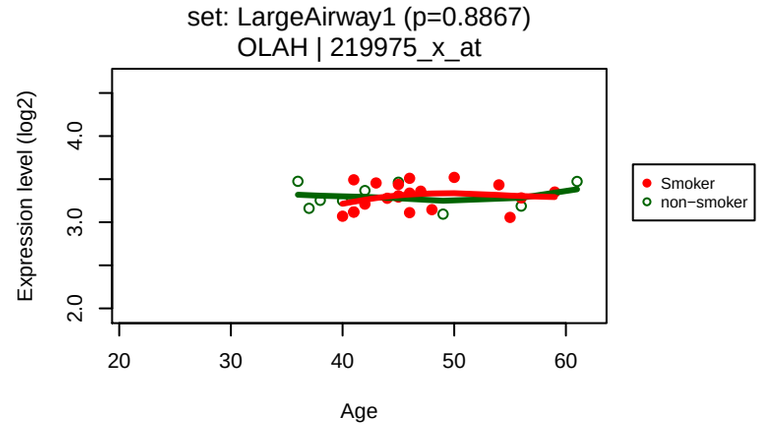

Gene expression of smokers and nonsmokers
gene: OLAH | 219975_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1611
p=0.6387
p=0.9408
p=0.8867
p=0.0081
p=0.2597
p=0.3971
p=0.453
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

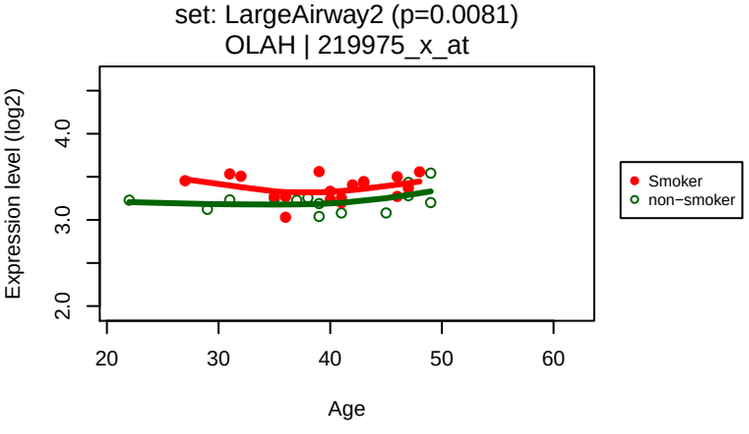

Gene expression ratio (smokers/nonsmokers)
gene: OLAH | 219975_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1611
p=0.6387
p=0.9408
p=0.8867
p=0.0081
p=0.2597
p=0.3971
p=0.453
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

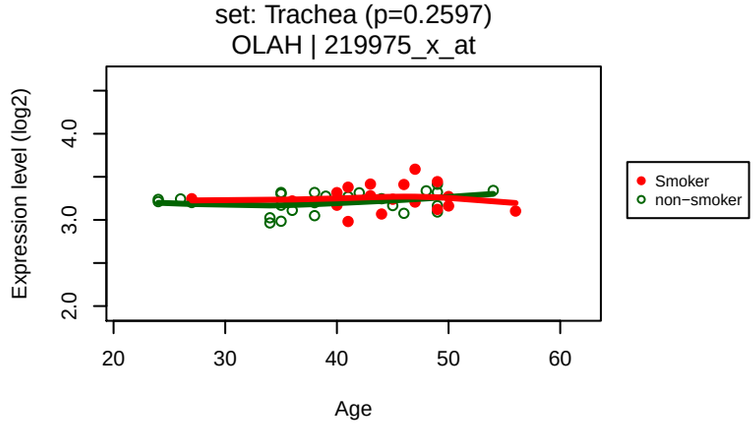

Gene expression ratio in Males
gene: OLAH | 219975_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5723
p=0.1364
p=0.4851
p=0.7983
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OLAH | 219975_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7424
p=0.0271
p=0.0501
p=0.2191
SmallAirway2
LargeAirway2
Trachea
Macrophage2

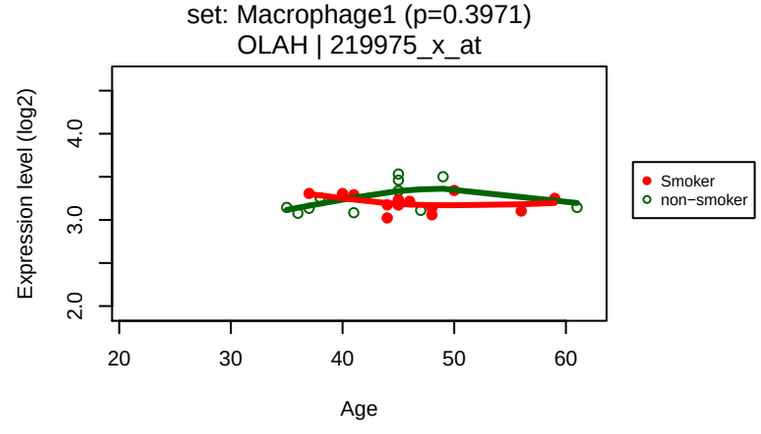

set: SmallAirway1 (p=0.1611)
OLAH | 219975_x_at
Expression level (log2)
Age
Smoker
non−smoker

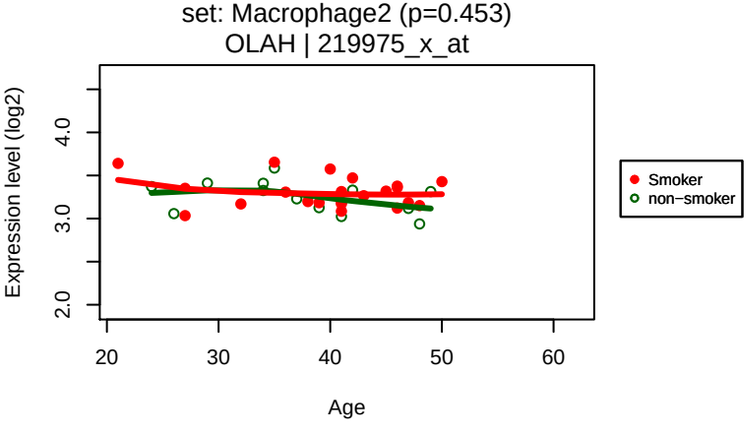

set: SmallAirway2 (p=0.6387)
OLAH | 219975_x_at
Expression level (log2)
Age
Smoker
non−smoker

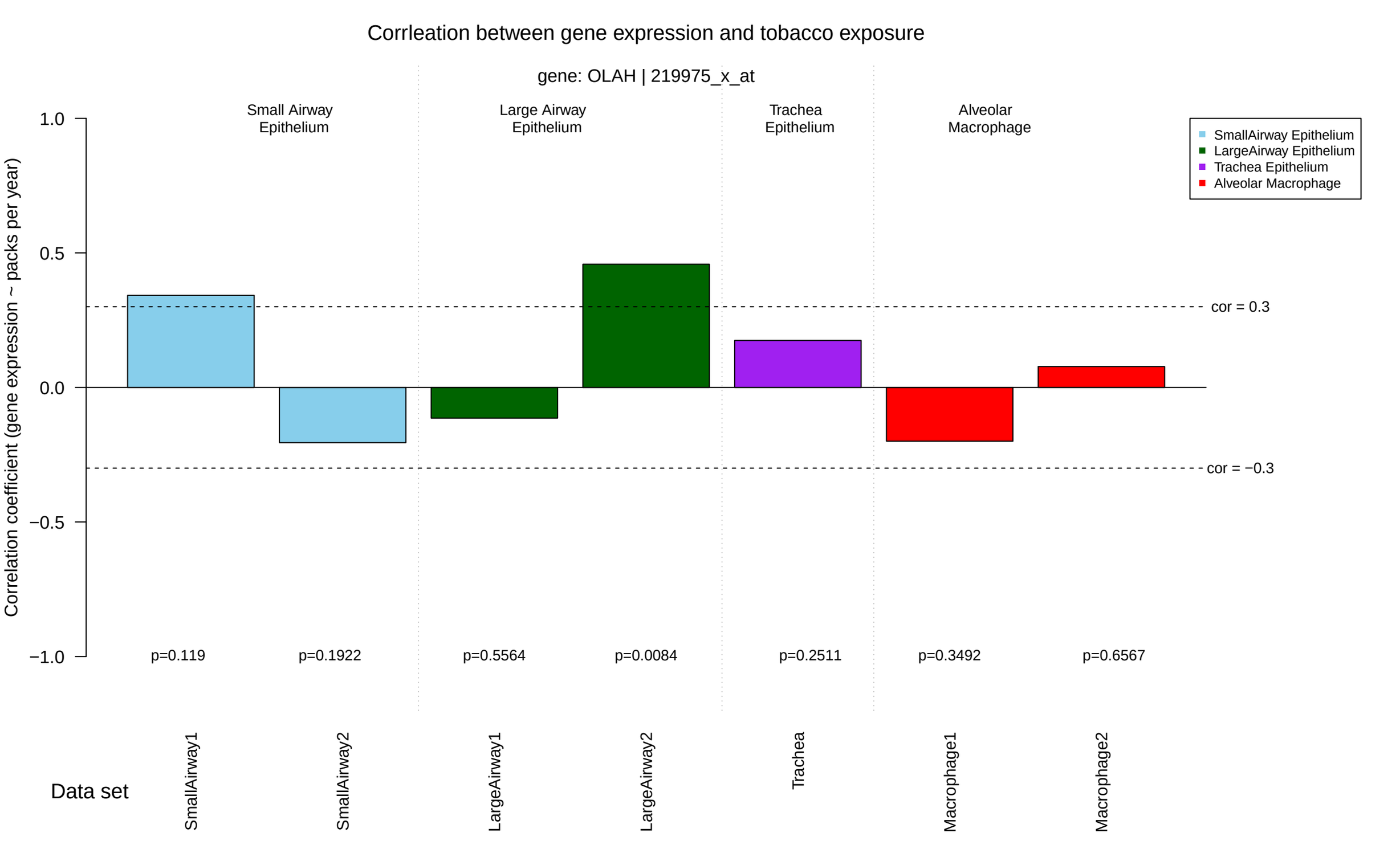

set: LargeAirway1 (p=0.8867)
OLAH | 219975_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0081)
OLAH | 219975_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2597)
OLAH | 219975_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3971)
OLAH | 219975_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.453)
OLAH | 219975_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OLAH | 219975_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.119
p=0.1922
p=0.5564
p=0.0084
p=0.2511
p=0.3492
p=0.6567
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2